

UCS413: NETWORK PROGRAMMING

L T P Cr
2 0 2 3.0

Course Objectives: The course introduces programming applications that use computer networks. The focus is on problem solving with emphasis on network programming. The operation and characteristics of major computer networks are studied because of their strong influence on programming interfaces (APIs) and application design.

OSI Model Introduction: OSI Model Layers Functions, TCP IP Stack real World Analogy, Data Encapsulation and Decapsulation – Introduction, Data Encapsulation, Data Decapsulation, Data Encapsulation and Decapsulation on Forwarding nodes, Multi Node Topology, Local and Remote Subnets, L3 Routing Information L3 Routes

Subnetting: Data Delivery, Mac and IP Address, Network ID, Broadcast Addresses, Max Value and Control Bits, IP Address Configuration, Point to Point Links Mask, Broadcast Addresses In detail

Layer 2 and Layer 3 Routing: Routing Introduction: Basics, Ethernet Header format, Layer 2 Routing, ARP Goals, ARP Standard Message Format, Address Resolution Protocol, Address Resolution Protocol Demonstration, Layer 2 Switch Concept and Functioning, Need of L3 Routes, Semantics of Layer 3 Routes, Routing table Look up, L3 Routing Topology, Layer 3 Operations, Loopback interfaces – Introduction, properties, Routing using Loopback IP Address as Destination Address

Broadcast Domain and Collision Domain: Introduction, Collision Domain reduction by L2 Switches, Broadcast Domain reduction by L3 router, Application Layer: Introduction, HTTP Server Design and Implementation from Scratch, HTTP Server Demonstration, HTTP Server Code Walk

LANs and VLANs: LANs and Use Cases, LANs Problems: Immobility and Security Issues, Introduction to VLANs: Access and Trunk Ports of L2 Switch, 802.1Q VLAN Header, VLAN Tagging Rules, VLAN L2 Routing Example, VLAN Benefits such as Segmentation, Resolve Thrashing, Reduced Broadcast Domain, Mobility

Router - VLAN Routing: Introduction and Problem Statement, Concept of SVIs, L3 Router Configuration for VLAN forwarding Router to VLAN Forwarding – Example, Inter VLAN Routing Basics and Routing methodology, Ping, Linux TCP dump utility: Capture packets, TCP Dump and ping assignment

Transport Layer: Introduction, Header Stacking, Transport Layer Port Numbers, System Call Interface

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.

Socket Programming: Introduction to Socket Programming, Server Designing, Accept system call, Select System Call Implementing Multiplexing with Accept & Select System Calls , TCP Server Example, TCP Server Design Observation, TCP Client Design and Implementation, TCP Server Client Demonstration, TCP Server With Multiplexing: High Level Design, Implementation and Demonstration

Domain Name Server (DNS): Introduction, DNS Architecture and Geographical Distribution, A Hierarchical and Decentralized System, Hosting your Own website, Website Domain Name and FQDN, Top Level Domain Servers Classification, DNS Resolver, DNS Query types: Recursive, Iterative and Reverse DNS Query

Packet Encapsulation: IP in IP Encapsulation – Introduction, Heterogeneous Networks, IPv6-in-IP Encapsulation Problem Statement and Solution, IP Encapsulation Problem Statement and Solution, IP Encapsulation Applications

Type Length Value: Introduction, Need of TLVs, Understanding TLVs, TLV Addressing the problem of Heterogeneity, TLV Addressing the problem of Software Upgrade, Data Structure – STREAMS, TLV De-Serialization using STREAMS

Self - Learning Content: Application Layer Protocols: Telnet, FTP, HTTP, SSL, IPFS, SMTP, POP3, IMAP, Client Server architecture, P2P architecture, Secure Socket Layer (SSL).

Course Learning Outcomes (CLOs) / Course Objectives (COs):

After the completion of the course, the student will be able to:

1. Analyse the requirements of a networked programming environment and identify the issues to be solved;
2. Create conceptual solutions to those issues and implement a programming solution;
3. Understand the key protocols that support the Internet;
4. Apply several common programming interfaces to network communication;
5. Apply advanced programming techniques such as Broadcasting, Multicasting

Text Books:

1. Harold E. R., Java Network Programming (2013), 4th ed. O'Reilly.
2. James F. K., Keith W. R., Computer Networking: A Top-Down Approach (2009) 6th Ed.

Reference Book:

1. Stevens W. R., Fenner B., Rudoff A.M., Unix Network Programming The Socket Networking API (2004) 3rd Ed, Pearson Education.

Evaluation Scheme

S.No.	Evaluation Elements	Weightage (%)
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